

ToolsEnabled: AI Agents Determine Minimal Pancyclic Graphs with Six Chords, Demonstrated on Erdős Problem #1016

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Working draft. Code, proofs and data: <https://github.com/JoshuaPinckard/erdos-1016-pancyclic>

Abstract. Open problems about extremal graphs often come down to finite cases that have to be settled by computer search, and each search has to be designed, written, run and checked before anyone can rely on it. We introduce agent research in ToolsEnabled, where a manager agent assigns lanes to worker agents that do this work, and other agents review what they produce. We demonstrate it on Erdős Problem #1016 (Bloom, n.d.), which asks for the fewest edges $m(n) = n + h(n)$ of a pancyclic graph on n vertices. Griffin (2013) determined $m(n)$ for $n \leq 37$, and White and Claude (2026) and Robinfxa (2026) published $m(38)$ to $m(41)$. The agents show that $h(n) = 6$ for every $41 \leq n \leq 67$, and that the largest n with a 6-chord pancyclic graph is 67 or lies in $\{72, \dots, 79, 81, 83, 84, 85\}$.

The agents reduce each graph, a cycle C_n with k added chords, to one of finitely many shapes. The upper bounds are explicit chord sets checked in Lean 4, and the lower bounds are exhaustive searches on a CPU and two GPUs, with every GPU result re-derived by a separate claim tool. Additionally other agents review the work as a referee would and attack the claim tool with forged results.

Problem. A pancyclic graph has a cycle of every length $3, \dots, n$, so it has a Hamilton cycle, and $h(n)$ is the fewest chords that make C_n pancyclic. Bondy (1971) stated $\log_2(n-1) - 1 \leq h(n) \leq \log_2 n + \log_* n + O(1)$. Erdős believed the upper bound is closer to the truth but could not prove that $h(n) - \log_2 n \rightarrow \infty$. We write t_k for the largest n with $h(n) \leq k$. Griffin’s table gives $t_1, \dots, t_4 = 5, 8, 14, 24$.

Theorem 1. $h(n) = 6$ for every $41 \leq n \leq 67$. Hence $m(n) = n + 6$ on this range, and $t_5 = 40$. For $41 \leq n \leq 67$ let F_n be C_n (vertices $0, \dots, n-1$) with the chords $\{0, 2\}$, $\{0, n-7\}$, $\{1, 13\}$, $\{3, n-6\}$, $\{4, 31\}$, $\{n-8, n-5\}$. Its cycle lengths are exactly $[3, 32] \cup [n-34, n]$, which is all of $[3, n]$ when $n \leq 67$, and F_{68} misses only the length 33. The Lean theorem `family_pancyclic_41_67` checks every case in the kernel and uses only the axioms `propext`, `Classical.choice` and `Quot.sound`. For the lower bound, an exhaustive shape search finds no pancyclic C_n with 5 chords at any n from 41 to 58, and no 5-chord shape can reach beyond $n = 58$. At $n = 41$ a complete GPU enumeration of all 5-chord sets agrees, which confirms $m(41) = 47$ (Robinfxa, 2026) by a second method.

Theorem 2. No 6-chord pancyclic graph exists for $n \in \{68, 69, 70, 71, 80, 82\}$ or for any $n \geq 86$. Hence $t_6 = 67$ or $t_6 \in \{72, \dots, 79, 81, 83, 84, 85\}$. A CPU branch and bound rules out every n from 88 to 111, and 111 is the largest n any 6-chord shape can reach. The GPU runs rule out 68 to 71, 80, 82 and 86 to 89, so 88 and 89 are ruled out by two different programs that share the same shape census. The runs stopped on 20 September, when levels 84 and 85 each lacked one tier. A blind control run over all of level 67 found F_{67} at its expected shape, and it also found a second 67-vertex graph that is not isomorphic to F_{67} , with the chords $\{0, 27\}$, $\{18, 30\}$, $\{28, 37\}$, $\{29, 37\}$, $\{31, 38\}$, $\{32, 39\}$.

Method. A cycle with k chords is a subdivision of one of finitely many multigraphs, the shapes, and there are 1,236 shapes for $k = 5$ and 21,878 for $k = 6$. Each cycle of the graph uses a fixed set of arcs, so its length is a sum of arc lengths. A shape with F distinct cycle forms can be pancyclic only if $n \leq F + 2$, which bounds every search. The CPU solver decides each pair of a shape and an n by branch and bound. The GPU pipeline enumerates the arc lengths directly, and in most tiers it prunes them with a pairwise interval-Hall test that is necessary for pancyclicity. A tier counts as done only when a separate tool re-derives all of its work units from the hashed census, and for the pruned tiers the tool also runs from a frozen copy of the code, rebuilds every pruning table and matches the composition counts shape by shape. The arithmetic of the prune and its Hall inequalities are proved in Lean, on the condition that the shape census is complete.

Agent process. The work ran from 13 to 20 September 2026 in one ToolsEnabled agent tree. The manager agent split the problem into lanes and merged the results. The worker agents searched the literature, built the shape reduction, the CUDA kernels and the Lean certificates, and ran the searches on an RTX 5060 Ti desktop and an RTX 4070 laptop. Other agents wrote a referee-style report that found overclaims in the draft and re-implemented the verifiers. They also attacked the claim tool with forged tiers, and two of the forgeries passed until the tool was fixed. The author set the problem, the direction and the compute limits, and checked the outputs.

Limitations. Every non-existence result rests on exhaustive computation and is not a Lean proof. The shape census is the one reduction step that is not checked in Lean, and the CPU and GPU checks both depend on it. Twelve levels were not finished, so t_6 is not yet known exactly. The values $m(38)$ to $m(41)$ were published before this work (White & Claude, 2026; Robinfxa, 2026), and the body of the chapter on minimal pancyclicity in George, Khodkar and Wallis (2016) was not obtained. The author of this paper is the founder of ToolsEnabled, Inc., generating a conflict of interest.

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